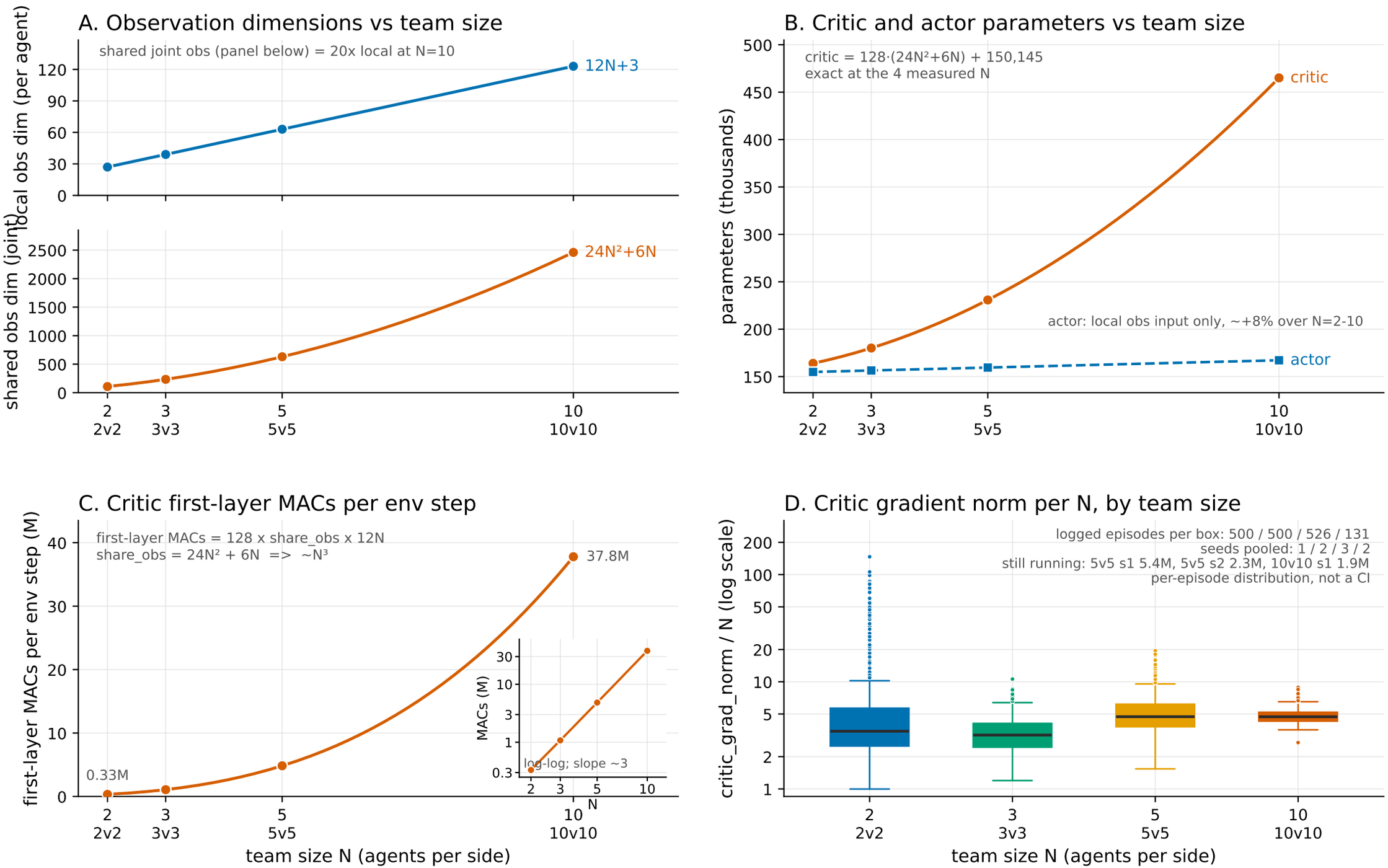

Panels A-C: critic_scaling.json measured rows only (N=2/3/5/10); the N=20/50/100 extrapolated rows are NOT plotted. Probe: one pass per model (cuda:0, batch_threads=12, batch_rows=12N); the closed forms are exact at the four measured N. N = agents per team; total agents = 2N. A: local and shared obs dims on separate y-axes; shared is 20x local at N=10. B: the critic head reads share_obs, the actor reads local obs. C: first-layer MACs (multiply-accumulates) per env step = 128 x share_obs x 12N, ~N^3 asymptotically (inset: log-log). D: normalized_scaling.csv rows labelled 'scaling noweapon' only (official 2v2 family excluded, not mixed); y = critic_grad_norm / N (per team size, not per agent) over 500/500/526/131 episodes pooled across 1/2/3/2 seeds; boxes = per-episode median / IQR / 1.5-IQR whiskers, dots beyond - an episode distribution, not a confidence interval; N=10 seed 1 and 5v5 seeds 1-2 were still running, so those boxes may shift. No smoothing applies (no time series). All panels are diagnostic: growth in dims / parameters / MACs is an architectural property and does NOT by itself show critic learning difficulty or a curse of multiagency; that claim needs controlled critic-variant experiments, not run this round.